

DA-WAM: DECISION-ALIGNED FUTURE LATENTS FOR DRIVING WORLD MODELS

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ABSTRACT

Anticipating how scenes evolve under ego actions is fundamental to safe autonomous driving, yet the full potential of world models for decision-making remains unrealized. The critical challenge lies in ensuring that future modeling is not merely predictive, but decision-informative: the predicted future must directly shape which trajectory is selected. Existing approaches decouple future representation learning from planning optimization, or share predicted states across trajectory candidates, thereby diluting the action-specific consequences that ought to guide selection. To bridge this gap, we propose DA-WAM, a framework that unifies predictive representation learning, action-conditioned future modeling, and trajectory scoring under a single decision-making objective. DA-WAM maintains predictive supervision throughout planner optimization via an online encoder and a stable momentum target, allowing future representations to co-evolve with the driving task. An action-conditioned predictor generates a distinct future latent state per trajectory candidate, which is then evaluated by a future-latent-conditioned factorized scorer. For the expert-matched trajectory, the predicted future latent is supervised by the observed future representation, while safety-critical hard negatives provide additional supervision near planning boundaries. Extensive experiments on NAVSIM-v1 and NAVSIM-v2 demonstrate state-of-the-art performance, while ablations and diagnostic analyses validate the key components. Code: <https://github.com/LeapWM/da-wam>.

1 INTRODUCTION

Safe and effective autonomous driving requires reasoning about how the scene would evolve under each candidate ego action. World models address this challenge by predicting future visual or latent states from current observations and proposed actions Wang et al. (2024); Li et al. (2025b); Zheng et al. (2025). Yet the fundamental question is not simply whether a model can predict the future, but whether its predictions are decision-informative: do the predicted consequences of each candidate action directly determine how that trajectory is evaluated? To realize the full potential of world models, future prediction must therefore be tightly coupled with decision making, such that each candidate is scored against the future predicted specifically for that action.

Existing approaches pursue future prediction along two broad directions, but neither direction alone guarantees this coupling. Methods such as DriveWorld Min et al. (2024), LAW Li et al. (2025b), Drive-JEPA Wang et al. (2026c), and Latent-WAM Wang et al. (2026b) use temporal prediction primarily to strengthen scene representations or trajectory learning. Although these approaches learn rich predictive features, predictive pretraining and planner optimization are often stage-separated; fixed or frozen encoders consequently cannot adapt their future representations to the specific demands of trajectory scoring. A second line of work brings predicted futures more directly into planning. Drive-WM Wang et al. (2024) generates future views for planning, WoTE Li et al. (2025c)

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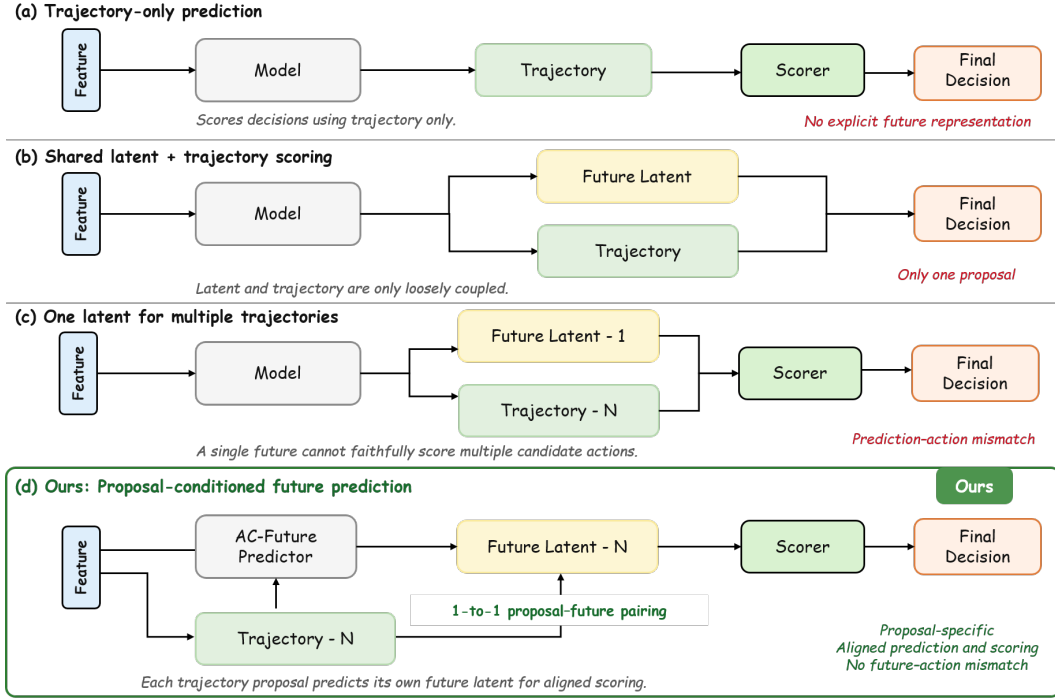


Figure 1: **Prediction-action alignment in trajectory scoring.** (a) Trajectory-only prediction provides no explicit future representation to the scorer. (b) Loosely coupled latent fusion incorporates a future representation but generates only a single trajectory proposal, precluding candidate-specific future comparison. (c) Sharing one future latent across multiple candidates creates a prediction-action mismatch. (d) DA-WAM predicts a distinct future latent for each candidate and scores the trajectory with its corresponding latent, establishing a one-to-one trajectory-future correspondence.

evaluates trajectories through a BEV world model, World4Drive Zheng et al. (2025) reasons over intention-conditioned latent futures, and DriveFuture Hong et al. (2026) conditions its planner on a predicted future latent. These methods move closer to using future predictions at decision time, yet predicted states may still be shared, pooled, or only weakly associated with individual candidates. As summarized in Fig. 1, these designs weaken the correspondence between an action and its predicted consequence. The scorer may therefore rely primarily on geometric cues rather than the scene-conditioned future content that distinguishes safe from unsafe outcomes.

We argue that the planning value of a world model is bounded by how directly its predictions influence candidate-level scoring. Evaluating each candidate against its own predicted future would allow the planner to exploit action-specific consequences, such as collisions, lane departures, or traffic-rule violations, when distinguishing geometrically similar yet safety-critical trajectories. Realizing this capability requires overcoming two interconnected barriers. At the representation level, predictive features must continue to adapt during planner optimization so that the learned future structure remains aligned with the planning objective. At the planning level, the scorer must evaluate a distinct future latent state for each candidate; shared or pooled futures obscure the action-specific consequences that should guide selection.

To this end, we propose DA-WAM, a decision-aligned world action model. DA-WAM maintains predictive supervision throughout planner optimization via a LoRA-adapted Video Joint Embedding Predictive Architecture (V-JEPA) 2.1 online encoder paired with an exponential moving average (EMA) target encoder, ensuring that future representations co-evolve with the driving task rather than being frozen after pretraining. An action-conditioned predictor generates a distinct future latent state for every trajectory candidate through explicit scene-trajectory interaction, and a factorized future-latent-conditioned scorer evaluates each candidate jointly with its corresponding latent state. Because offline logs provide observed futures only for the executed expert trajectory, dense JEPA supervision is applied to the expert-matched candidate, while safety-critical hard negatives supply

additional local comparisons near planning boundaries. These hard negatives are geometrically similar to the expert-matched candidate but differ in safety outcomes, discouraging the scorer from relying on geometry alone.

Our contributions are threefold:

- We propose decision-aligned future latent learning, which associates each candidate trajectory with a distinct predicted future and uses its action-specific consequences to guide trajectory selection.
- We introduce a unified training framework that continues predictive representation learning during planner optimization, allowing the latent space to adapt to the driving objective rather than remain fixed after pretraining.
- We provide a supervision strategy that combines expert-matched future targets with safety-critical hard negatives, improving the scorer’s ability to distinguish geometrically similar candidates that lead to different safety outcomes.

2 RELATED WORK

2.1 JOINT-EMBEDDING PREDICTION AND LATENT PLANNING

JEPAs capture high-level semantic and temporal representations by predicting latent features of future states rather than reconstructing pixel-level details Assran et al. (2025); Mur-Labadia et al. (2026). In end-to-end autonomous driving, this paradigm offers an efficient avenue to model scene evolution, which is crucial for forecasting the downstream impact of ego decisions. Recent works such as DriveWorld Min et al. (2024) and LAW Li et al. (2025b) use future dynamics to enrich visual scene representations for motion planning. Drive-JEPA Wang et al. (2026c) adapts pretrained video JEPAs to trajectory planning via fine-tuning, while Auto-JEPA Yang et al. (2026) predicts continuous intent embeddings with a frozen visual backbone to rank candidate paths. Similarly, Latent-WAM Wang et al. (2026b) jointly trains a spatial encoder against an exponential moving average latent target but discards the dynamics branch at test time.

A fundamental tension underlying these approaches is the trade-off between the generality of predictive representations and the task-specific needs of trajectory scoring. Existing methods typically address this trade-off through frozen pretraining, multistage pipelines, or inference-time removal, which can weaken the coupling between future dynamics and policy optimization. In contrast, we maintain predictive JEPA supervision throughout planner optimization. This enables future latent representations to co-evolve with the scoring objective and serve as direct conditioning inputs during inference rather than auxiliary training-time features.

2.2 ACTION-CONDITIONED DRIVING WORLD MODELS

Recognizing that visual environments evolve differently under different ego maneuvers, recent world models explicitly condition future predictions on hypothetical actions or trajectory proposals. Drive-WM Wang et al. (2024) synthesizes multiview video futures under alternative driving commands, whereas LAW Li et al. (2025b) and WoTE Li et al. (2025c) predict trajectory-conditioned latent states or bird’s-eye-view (BEV) dynamics. World4Drive Zheng et al. (2025) forecasts multiple intention-guided latent futures and validates paths using an internal evaluator. Concurrently, Drive-Future Hong et al. (2026), IDOL Zhang et al. (2026), and LCDrive Tan et al. (2026) use future latents for diffusion guidance, inverse-dynamics refinement, and latent chain-of-thought reasoning, respectively.

While these works validate the premise of action conditioning, they often aggregate, pool, or weakly fuse the predicted futures across proposals. Consequently, the candidate evaluator receives a homogenized scene representation, diluting the fine-grained, safety-critical consequences specific to each trajectory. We bridge this gap by establishing an explicit one-to-one correspondence: DA-WAM generates a distinct future latent state for every candidate trajectory and feeds it directly into the scorer, allowing trajectory selection to use candidate-specific counterfactual evidence.

2.3 CANDIDATE GENERATION AND TRAJECTORY SCORING

Candidate-based planners generate multiple plausible trajectories and select the optimal one according to scene context and planning objectives. DiffusionDrive efficiently models multimodal action distributions through anchor-guided truncated diffusion Liao et al. (2025). DrivoR compresses multi-camera features into camera-aware register tokens and uses separate transformer decoders to generate and score candidates Kirby et al. (2026). GTRS and ZTRS formulate candidate evaluation as explicit trajectory scoring Li et al. (2025e;d). Beyond geometric and scene-based scoring, DriveSuprim employs coarse-to-fine trajectory selection to distinguish hard-negative trajectories Yao et al. (2026), and BeyondDrive constructs safety-critical, expert-proximate hard negatives to learn safety boundaries in trajectory space Wang et al. (2026a).

These works have significantly advanced both the generation and evaluation of trajectory candidates. Nonetheless, the scoring signal remains predominantly anchored in current scene geometry and immediate motion patterns. Predicted futures are rarely incorporated into the scorer in an explicit, per-candidate manner, despite their potential to distinguish geometrically similar but safety-critical candidates. We operate within the established candidate-based planning paradigm and augment the scorer with action-conditioned future latent states, thereby enabling trajectory selection based on predicted consequences rather than current geometry alone.

3 METHODOLOGY

3.1 PROBLEM FORMULATION AND FRAMEWORK OVERVIEW

In this section, we formalize the trajectory planning problem and describe how DA-WAM couples future latent prediction with per-candidate trajectory evaluation.

Given the current visual observation X_t , the planner first obtains a set of N ego trajectory candidates $\mathcal{T} = \{\tau_i\}_{i=1}^N$. The planning objective is to evaluate the utility of each candidate and select the trajectory with the optimal predicted outcome.

The overall architecture of DA-WAM is illustrated in Fig. 2. Given X_t and $\mathcal{T}$, the framework operates in three main steps:

- **Observation Encoding:** The online encoder E_θ maps X_t to scene latent tokens Z_t . During training, an EMA target encoder $E_{\bar{\theta}}$ additionally extracts target features $Z_{t+\Delta}$ from the future frame $X_{t+\Delta}$.
- **Action-Conditioned Future Prediction:** Each trajectory τ_i is mapped to an action representation a_i . A shared predictor P_ϕ then fuses a_i with Z_t to forecast the candidate-specific future latent state $\hat{Z}_i$.
- **Trajectory Scoring:** A shared scorer S_ψ evaluates the triplet $(Z_t, a_i, \hat{Z}_i)$ for each candidate, predicting both interpretable planning factors $\hat{\mathbf{q}}_i$ and an overall utility score $\hat{s}_i$. The candidate with the highest score is selected for execution.

Training and Inference Paradigm. Training accounts for the fundamental counterfactual limitation of offline driving logs: each scene provides only an expert trajectory τ^{exp} and its corresponding executed future $X_{t+\Delta}$. To prevent assigning the observed expert future to unexecuted counterfactual trajectories, we match τ^{exp} to the closest candidate and apply the dense latent prediction loss exclusively to its predicted future. Meanwhile, all candidates receive factor, utility, and ranking supervision, with expert-proximate hard negatives providing critical contrastive signals near planning decision boundaries. During inference, DA-WAM operates without access to future observations or expert priors, and the EMA target network is omitted. Only the online encoder, predictor, and scorer are activated to evaluate generated candidates in real time. The main notation used throughout the method is summarized in Appendix A.

3.2 JEPA-DRIVEN PREDICTIVE REPRESENTATION ADAPTATION

To preserve predictive world knowledge while tailoring representations to downstream navigation, DA-WAM adapts features via a dual online-target architecture. Given the current observation X_t ,

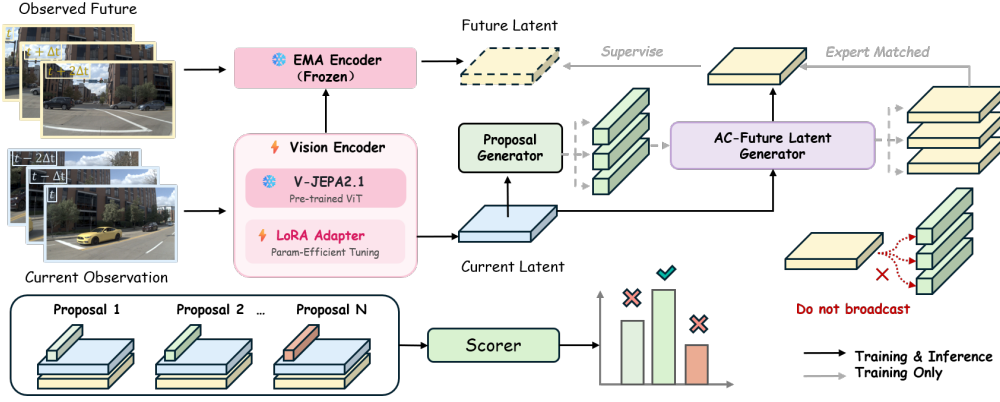


Figure 2: **Overview of DA-WAM.** The online encoder E_θ first maps the current observation X_t to scene tokens Z_t . For each candidate trajectory τ_i , its action representation a_i is combined with Z_t via the predictor P_ϕ to forecast a candidate-specific future latent state $\hat{Z}_i$. A shared scorer S_ψ then evaluates the triplet $(Z_t, a_i, \hat{Z}_i)$ to predict interpretable planning factors $\hat{q}_i$ and an overall utility score $\hat{s}_i$. **Training:** An EMA target encoder extracts target latents $Z_{t+\Delta}$ from the observed future frame to supervise only the expert-matched prediction, while safety-critical hard negatives enhance boundary discrimination. **Inference:** Only the online encoder, predictor, and scorer are activated, requiring no future observations or expert priors.

the online encoder extracts spatial scene tokens:

$$Z_t = E_\theta(X_t), \quad (1)$$

where $Z_t \in \mathbb{R}^{M \times D}$ comprises M latent tokens of feature dimension D .

We initialize E_θ with a pretrained V-JEPA 2.1 backbone and inject Low-Rank Adaptation (LoRA) modules into selected transformer layers. While the base network remains frozen, the LoRA parameters are jointly updated by gradients from future prediction and trajectory planning. This design retains the pretrained backbone’s representational capabilities while adapting the latent space to driving-specific objectives.

During training, the observed future frame $X_{t+\Delta}$ is processed by a target network with stop-gradient (sg):

$$Z_{t+\Delta} = \text{sg}(E_{\bar{\theta}}(X_{t+\Delta})), \quad (2)$$

whose parameters $\bar{\theta}$ are updated as an EMA of the online parameters θ :

$$\bar{\theta} \leftarrow \mu \bar{\theta} + (1 - \mu) \theta, \quad (3)$$

where $\mu \in [0, 1)$ is the momentum coefficient. This momentum update yields stable, slowly-evolving regression targets and prevents representational collapse. The target branch is used exclusively during training; only the adapted online encoder is deployed for inference.

3.3 ACTION-CONDITIONED COUNTERFACTUAL WORLD MODELING

Autonomous planning requires anticipating outcomes conditioned on specific ego actions. Rather than assuming a single global future, DA-WAM constructs candidate-specific future latent states for all N candidate trajectories.

Each candidate τ_i is parameterized as a temporal sequence of future ego states (e.g., position and heading) and encoded as an action representation by a trajectory encoder:

$$a_i = E_\tau(\tau_i). \quad (4)$$

A shared future predictor P_ϕ then uses the action query to attend to the current scene tokens:

$$\hat{Z}_i = P_\phi(Q = a_i, K = Z_t, V = Z_t), \quad i = 1, \dots, N. \quad (5)$$

Within P_ϕ , a_i serves as the query, while the spatial scene tokens Z_t provide the keys and values. The resulting action-specific context conditions the latent prediction tokens, enabling a single observation to produce distinct counterfactual latent futures. Sharing predictor parameters across all N candidates avoids introducing candidate-specific model biases; differences among $\hat{Z}_i$ are instead driven by the action queries a_i .

Expert Matching for Counterfactual Futures. Although the model predicts N counterfactual futures, offline datasets record only the outcome corresponding to the executed expert trajectory τ^{exp} . Applying the observed future target to unexecuted candidates would provide incorrect supervision for counterfactual actions. Therefore, we restrict dense predictive supervision to the expert-matched candidate:

$$i^{\text{exp}} = \arg \min_i \text{ADE}(\tau_i, \tau^{\text{exp}}), \quad (6)$$

where $\text{ADE}(\cdot)$ denotes average displacement error.

The predictive loss is then computed token-wise exclusively for the expert-matched candidate:

$$\mathcal{L}_{\text{pred}} = \frac{1}{M} \sum_{m=1}^M \ell(\hat{Z}_{i^{\text{exp}}, m}, Z_{t+\Delta, m}), \quad (7)$$

where $\ell(\cdot)$ is the adopted feature regression loss.

The remaining $N - 1$ counterfactual latents cannot receive direct feature-level supervision because their corresponding outcomes are unobserved. Instead, they are optimized indirectly through the downstream trajectory-scoring losses. This alignment avoids assigning observed outcomes to unexecuted actions while still allowing all predicted latents to inform final trajectory selection.

3.4 FUTURE-LATENT-CONDITIONED TRAJECTORY SCORING

Predicting action-conditioned future states is valuable only if these imagined outcomes directly govern trajectory decision-making. To this end, DA-WAM evaluates each candidate trajectory by explicitly conditioning its score on its own predicted future latent state.

For each candidate trajectory τ_i , a scoring transformer cross-attends the current scene tokens Z_t , the action representation a_i , and the predicted future latent state $\hat{Z}_i$ to produce a unified trajectory representation:

$$h_i = S_\psi^{\text{enc}}(Z_t, \hat{Z}_i, a_i), \quad (8)$$

where ψ encompasses all learnable parameters in the scoring module. The encoder S_ψ^{enc} preserves fine-grained token-level interactions rather than pooling futures into a coarse proposal-invariant vector. Because its parameters are shared across candidates, differences in scores arise from candidate geometry and the corresponding predicted outcomes rather than candidate-specific scorer parameters.

Factorized Planning Heads. To ground trajectory evaluation in explicit driving priors, dedicated linear heads decode intermediate planning-relevant factors directly from h_i :

$$\hat{\mathbf{q}}_i = S_\psi^{\text{factor}}(h_i) = [\hat{q}_i^{\text{NC}}, \hat{q}_i^{\text{DAC}}, \hat{q}_i^{\text{EP}}, \hat{q}_i^{\text{TTC}}, \hat{q}_i^{\text{Comfort}}]. \quad (9)$$

These entries represent no-at-fault collision (NC), drivable area compliance (DAC), ego progress (EP), time to collision (TTC), and comfort, respectively. Each dimension is supervised by simulation-derived or rule-based trajectory metrics.

Subsequently, a utility head aggregates the holistic feature h_i and the predicted factor vector $\hat{\mathbf{q}}_i$ to output a comprehensive scalar utility:

$$\hat{s}_i = S_\psi^{\text{score}}(h_i, \hat{\mathbf{q}}_i), \quad (10)$$

where $\hat{s}_i$ serves as the final ranking score for selecting a candidate during deployment. This factorized architecture improves interpretability while regularizing the trajectory feature space.

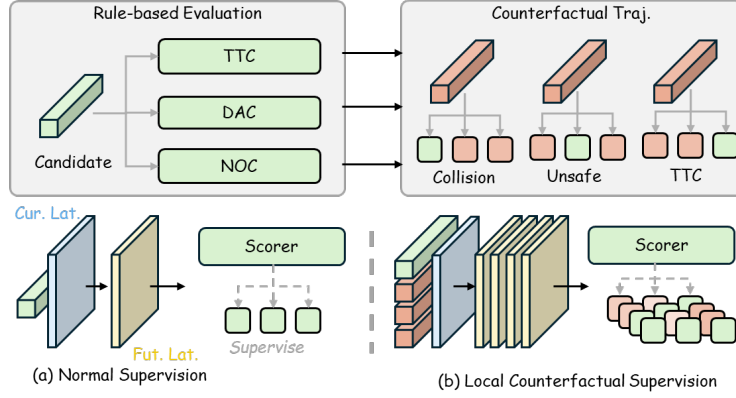


Figure 3: **Safety-critical hard-negative trajectory supervision.** Conventional training evaluates a sparse set of generated candidates using rule-based NC, DAC, and TTC factors. DA-WAM additionally retrieves expert-proximate hard-negative trajectories that are geometrically similar to the expert trajectory but differ in safety outcomes. Generated candidates and hard negatives query the same scene representation and share one future-latent-conditioned trajectory scorer. Hard-negative labels are training-only planning targets rather than observed future representations.

Trajectory-Level Counterfactual Safety Supervision. Randomly sampled candidate sets often exhibit large geometric differences, allowing the scorer to rely on coarse cues such as curvature and speed rather than scene-dependent safety consequences. We therefore augment each scene with expert-proximate, safety-critical hard negatives. These trajectories remain geometrically close to the expert but lead to substantially different safety outcomes, providing counterfactual supervision near planning boundaries.

For each scenario, candidate trajectories τ_j^- are retrieved from an offline trajectory bank subject to dual constraints:

$$\begin{aligned} d_{\text{traj}}(\tau_j^-, \tau^{\text{exp}}) &< \epsilon_{\text{geo}}, \\ \Delta_{\text{safety}}(\tau_j^-, \tau^{\text{exp}}) &> \epsilon_{\text{safety}}, \end{aligned} \quad (11)$$

where ϵ_{geo} enforces geometric closeness to the expert, while ϵ_{safety} requires a pronounced degradation in safety metrics (e.g., an impending collision or lane departure).

Figure 3 summarizes how these trajectories augment the generated candidates during training. Each τ_j^- is appended to the candidate set, encoded as $a_j^- = E_{\tau}(\tau_j^-)$, and processed by Eq. 5 to condition its own future latent before entering the shared scorer. Because its corresponding visual future is unobserved, a hard negative is excluded from expert matching and direct future-feature supervision, but still receives factor, utility, and ranking targets. This construction encourages the scorer to distinguish the consequences of different ego behaviors under the same scene context.

3.5 TRAINING OBJECTIVES AND INFERENCE

The proposed framework is trained end-to-end using a composite loss function derived from predictive feature alignment and planning objectives.

For each candidate i , we use external planning metrics to provide factor targets $\mathbf{q}_i$ and an overall utility target s_i . The factorized planning loss enforces fidelity to specific driving requirements:

$$\mathcal{L}_{\text{factor}} = \sum_i \sum_{k \in \mathcal{K}} \lambda_k \ell_k(\hat{q}_i^k, q_i^k), \quad (12)$$

where $\mathcal{K}$ denotes the set of planning factors, such as no-at-fault collision and time to collision. The loss function ℓ_k is tailored to the target q_i^k ; for example, mean squared error (MSE) is used for continuous factors, while binary cross-entropy (BCE) is used for binary factors.

The direct supervision on the final utility score is given by:

$$\mathcal{L}_{\text{score}} = \sum_i \ell_{\text{score}}(\hat{s}_i, s_i). \quad (13)$$

To ensure robust relative ranking, we construct preference pairs (i, j) based on their ground-truth utilities s_i and s_j :

$$y_{ij} = \mathbb{I}[s_i > s_j]. \quad (14)$$

The pairwise ranking objective employs a standard cross-entropy formulation over the sigmoid difference of predicted scores:

$$\mathcal{L}_{\text{rank}} = - \sum_{(i,j)} [y_{ij} \log \sigma(\hat{s}_i - \hat{s}_j) + (1 - y_{ij}) \log \sigma(\hat{s}_j - \hat{s}_i)]. \quad (15)$$

As noted in Section 3.4, pairs involving safety-critical hard negatives are either oversampled or assigned greater loss weights, emphasizing preferences between safe and unsafe candidates near local decision boundaries.

The total training objective integrates the predictive feature-alignment loss ($\mathcal{L}_{\text{pred}}$ from Eq. 7) with the planning losses:

$$\mathcal{L} = \lambda_{\text{pred}} \mathcal{L}_{\text{pred}} + \lambda_{\text{factor}} \mathcal{L}_{\text{factor}} + \lambda_{\text{score}} \mathcal{L}_{\text{score}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}}. \quad (16)$$

This composite objective respects the observational constraints of offline data: $\mathcal{L}_{\text{pred}}$ is applied exclusively to the expert-matched candidate, which provides observed-future supervision, while all candidates, including generated candidates and hard negatives, contribute to $\mathcal{L}_{\text{factor}}$, $\mathcal{L}_{\text{score}}$, and $\mathcal{L}_{\text{rank}}$.

Inference. During inference, the system requires only the current observation X_t and the set of generated trajectory candidates $\mathcal{T}$. Training-only components, including the target encoder $E_{\bar{\theta}}$, expert matching, and hard-negative retrieval, are not used.

The online inference sequence is as follows:

- The online encoder E_{θ} extracts the scene representation Z_t .
- The action-conditioned predictor P_{ϕ} generates a future latent state $\hat{Z}_i$ for every candidate $\tau_i \in \mathcal{T}$.
- The future-latent-conditioned scorer S_{ψ} computes the planning factors $\hat{\mathbf{q}}_i$ and overall utility $\hat{s}_i$ for each candidate based on $(Z_t, \hat{Z}_i, a_i)$.

The final planned trajectory τ^* is selected by maximizing the predicted utility score:

$$\tau^* = \arg \max_{\tau_i \in \mathcal{T}} \hat{s}_i. \quad (17)$$

At inference, each candidate is evaluated together with its own predicted future latent, preserving the one-to-one correspondence between candidate trajectories and predicted outcomes during ranking.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets and Evaluation Metrics. Our primary evaluation is conducted on the NAVSIM-v1 `navtest` split Dauner et al. (2024), which contains 12,146 driving scenarios. We report the Predictive Driver Model Score (PDMS) together with its five components: No-at-Fault Collision (NC), Drivable Area Compliance (DAC), Ego Progress (EP), Time to Collision (TTC), and Comfort. Drivable Direction Compliance (DDC), provided by the evaluation pipeline, is included as an additional diagnostic. Unless stated otherwise, all scores are multiplied by 100. We further evaluate DA-WAM under the broader compliance criteria of NAVSIM-v2 `navtest`, using the Extended Predictive Driver Model Score (EPDMS).

Table 1: **NAVSIM-v1 benchmark comparison.** Camera-only methods on the `navtest` split. All scores are scaled by 100, and higher is better for every metric.

Method	Venue	NC	DAC	TTC	Comfort	EP	PDMS
PDM-Closed Dauner et al. (2023)	CoRL’23	94.6	99.8	89.9	86.9	99.9	89.1
Human driver Dauner et al. (2024)	NeurIPS’24	100.0	100.0	100.0	99.9	87.5	94.8
Ego-stat. MLP Dauner et al. (2024)	NeurIPS’24	93.0	77.3	83.6	100.0	62.8	65.6
UniVLA Wang et al. (2026d)	ICLR’26	96.9	91.1	91.7	96.7	76.8	81.7
DrivingGPT Chen et al. (2025)	ICCV’25	98.9	90.7	94.9	95.6	79.7	82.4
UniAD Hu et al. (2023)	CVPR’23	97.8	91.9	92.9	100.0	78.8	83.4
DriveX-S Shi et al. (2025)	ICCV’25	97.5	94.0	93.0	100.0	79.7	84.5
World4Drive Zheng et al. (2025)	ICCV’25	97.4	94.3	92.8	100.0	79.9	85.1
VAD-v2 Jiang et al. (2026)	ICLR’26	98.1	94.8	94.3	100.0	80.6	86.2
PRIX Wozniak et al. (2026)	RA-L’26	98.1	96.3	94.1	100.0	82.3	87.8
DiffusionDrive Liao et al. (2025)	CVPR’25	98.2	96.2	94.7	100.0	82.2	88.1
DIVER Song et al. (2026)	TPAMI’26	98.5	96.5	94.9	100.0	82.6	88.3
AutoVLA Zhou et al. (2026)	NeurIPS’25	98.4	95.6	98.0	99.9	81.9	89.1
DriveVLA-W0 Li et al. (2026)	ICLR’26	98.7	99.1	95.3	99.3	83.3	90.2
ReCogDrive Xiong et al. (2026)	ICLR’26	97.9	97.3	94.9	100.0	87.3	90.8
Hydra-MDP++ Li et al. (2025a)	arXiv’25	98.6	98.6	95.1	100.0	85.7	91.0
DiffusionDriveV2 Zou et al. (2025)	arXiv’25	98.3	97.9	94.8	99.9	87.5	91.2
iPad Guo et al. (2026)	arXiv’25	98.6	98.3	94.9	100.0	88.0	91.7
SparseDriveV2 Sun et al. (2026)	arXiv’26	98.5	98.4	95.0	99.9	88.6	92.0
Centaur Sima et al. (2025)	arXiv’25	99.5	98.9	98.0	100.0	85.9	92.6
DrivoR Kirby et al. (2026)	CVPR’26	98.9	98.3	96.2	100.0	89.1	93.1
DriveSuprim Yao et al. (2026)	AAAI’26	98.6	98.6	95.5	100.0	91.3	93.5
DA-WAM	–	99.1	98.9	96.8	99.8	90.0	93.7

Implementation Details. Each input comprises two historical frames from the front camera. The proposal module produces 32 candidate trajectories, with each candidate represented by eight future ego poses. Conditioned on each candidate, the action-conditioned predictor forecasts a candidate-specific scene latent 0.5 seconds into the future. The visual encoders are initialized from pretrained V-JEPA 2.1 Mur-Labadia et al. (2026); the online branch is adapted with Low-Rank Adaptation (LoRA), and the target branch is updated by exponential moving average (EMA).

We train all primary NAVSIM-v1 variants for 20 epochs on 8 GPUs with a batch size of 8 per GPU, selecting checkpoints according to validation performance. In matched studies, the training data, parameter initialization, proposal generator, optimization schedule, checkpoint-selection rule, and evaluation protocol are held fixed. The resulting controls compare four prediction settings: no future prediction, a shared global future, the current latent, and an action-conditioned future. We also isolate the contribution of hard-negative supervision.

4.2 MAIN RESULTS

Tables 1 and 2 summarize the public benchmark results. Among the compared learning-based planners, DA-WAM obtains the best overall planning score on both benchmarks, reaching 93.7 PDMS on NAVSIM-v1 and 87.7 EPDMS on NAVSIM-v2. We next examine the results under each evaluation protocol.

Benchmarking on NAVSIM-v1. Table 1 compares DA-WAM with camera-only methods on the NAVSIM-v1 `navtest` split. DA-WAM slightly surpasses the strongest prior learned planner in PDMS and achieves 99.1 NC, 98.9 DAC, and 90.0 EP. The result reflects a favorable balance between safety, road compliance, and driving progress. This public benchmark comparison complements the controlled matched study in Table 3.

Benchmarking on NAVSIM-v2. Table 2 reports results on the NAVSIM-v2 `navtest` leaderboard for methods using ResNet-34 and ViT/L backbones. Under the expanded metric set, DA-WAM achieves particularly strong TTC and Lane Keeping scores of 97.9 and 97.6, respectively. These results raise EPDMS to 87.7, exceeding the strongest comparison by 0.2 points.

Table 2: **NAVSIM-v2 benchmark comparison.** Methods with ResNet-34 and ViT/L visual backbones on the navtest split. All scores are scaled by 100, and higher is better for every metric.

Method	Img. Backbone	NC	DAC	DDC	TL	EP	TTC	LK	HC	EC	EPDMS
Ego Status MLP	ResNet-34	93.1	77.9	92.7	99.6	86.0	91.5	89.4	98.3	85.4	64.0
TransFuser Chitta et al. (2022)	ResNet-34	96.9	89.9	97.8	99.7	87.1	95.4	92.7	98.3	87.2	76.7
Hydra-MDP++ Li et al. (2025a)	ResNet-34	97.2	97.5	99.4	99.6	83.1	96.5	94.4	98.2	70.9	81.4
DriveSuprim Yao et al. (2026)	ResNet-34	97.5	96.5	99.4	99.6	88.4	96.6	95.5	98.3	77.0	83.1
ARTEMIS Feng et al. (2025)	ResNet-34	98.3	95.1	98.6	99.8	81.5	97.4	96.5	98.3	98.3	83.1
DiffusionDriveV2 Zou et al. (2025)	ResNet-34	97.7	96.6	99.2	99.8	88.9	97.2	96.0	97.8	91.0	87.5
SparseDriveV2 Sun et al. (2026)	ResNet-34	98.1	98.1	99.6	99.8	91.1	97.3	96.9	98.2	78.4	86.7
Hydra-MDP++ Li et al. (2025a)	ViT/L	98.4	98.0	99.4	99.8	87.5	97.7	95.3	98.3	77.4	85.1
DriveSuprim Yao et al. (2026)	ViT/L	97.8	97.9	99.5	99.9	90.6	97.1	96.6	98.3	77.9	86.0
DA-WAM	ViT/L	98.4	98.4	99.1	99.9	88.6	97.9	97.6	97.8	79.6	87.7

4.3 ABLATION STUDIES

Future-Prediction Configuration Ablation. Table 3 compares matched future-prediction configurations and evaluates the contribution of safety-critical hard-negative supervision.

Table 3: Matched ablation of future-prediction configurations on the NAVSIM-v1 navtest split. All metrics are scaled by 100, and higher is better. In the hard-negative column, a checkmark and a cross denote enabled and disabled supervision, respectively; a dash denotes that the setting is not applicable.

Configuration	Hard neg.	PDMS	NC	DAC	EP	TTC	Comfort
No Future Prediction	–	93.31	98.45	98.27	91.36	95.48	99.99
Shared Global Future	–	92.81	99.02	98.46	88.68	96.54	99.99
Current-Latent Conditioning	–	93.25	98.44	98.19	91.38	95.49	99.94
Action-Conditioned Future	✗	93.46	98.88	98.58	90.47	96.33	99.69
	✓	93.68	99.11	98.88	89.97	96.81	99.77

The no-future-prediction planner already achieves 93.31 PDMS, demonstrating the strength of a conventional end-to-end planner. The current-latent control performs similarly at 93.25, showing that an additional pathway alone provides little benefit. The shared-global-future control improves NC and TTC but reduces EP from 91.36 to 88.68, resulting in the lowest PDMS of 92.81. Sharing one future across all candidates introduces a prediction–action mismatch and encourages an averaged representation that weakens candidate discrimination.

Action conditioning restores the correspondence between each trajectory and its predicted future, raising PDMS to 93.46 without hard negatives. This exceeds the no-future-prediction, shared-global-future, and current-latent controls by 0.15, 0.65, and 0.21 points, respectively. Adding counterfactual safety supervision further improves PDMS to 93.68, together with higher NC, DAC, TTC, and Comfort, while EP decreases from 90.47 to 89.97. These results suggest that candidate-specific futures provide useful action-level evidence, while safety-critical hard negatives further sharpen the scorer’s discrimination near planning boundaries.

Predictive-Representation Ablation. This ablation jointly analyzes online-encoder adaptation, the V-JEPA predictive objective, and the target-encoder policy. Table 4 distinguishes V-JEPA 2.0 from V-JEPA 2.1, whose predictive objective uses dense latent supervision, while comparing frozen, LoRA-adapted, and fully fine-tuned online encoders. It then fixes the online encoder to LoRA and enables the V-JEPA 2.1 dense loss to compare frozen, separate, shared, and EMA target-encoder policies. All matched variants use the same initialization, proposal set, training schedule, and checkpoint-selection rule.

Table 4: Ablation of online-encoder adaptation, dense prediction loss, and target-encoder policy on the NAVSIM-v1 `navtest` split. In the Dense loss column, a checkmark denotes the V-JEPA 2.1 dense latent objective, whereas a cross denotes the V-JEPA 2.0 objective. PDMS is scaled by 100, and higher is better.

Adaptation	Dense loss	Target	PDMS
Online-encoder adaptation and predictive objective			
Frozen	✗	Frozen	91.26
Frozen	✓	Frozen	91.95
LoRA	✗	Frozen	92.74
LoRA	✓	Frozen	92.98
Full ft.	✓	Frozen	92.62
Target-encoder policy (LoRA + dense loss)			
LoRA	✓	Separate	93.10
LoRA	✓	Shared	93.34
LoRA	✓	EMA	93.68

Dense latent supervision improves PDMS for both the frozen encoder (91.26 to 91.95) and the LoRA-adapted encoder (92.74 to 92.98). Under the dense objective, LoRA adaptation outperforms full fine-tuning by 0.36 points. With LoRA fixed, the EMA target encoder achieves the best result, improving PDMS from 92.98 with a frozen target to 93.68.

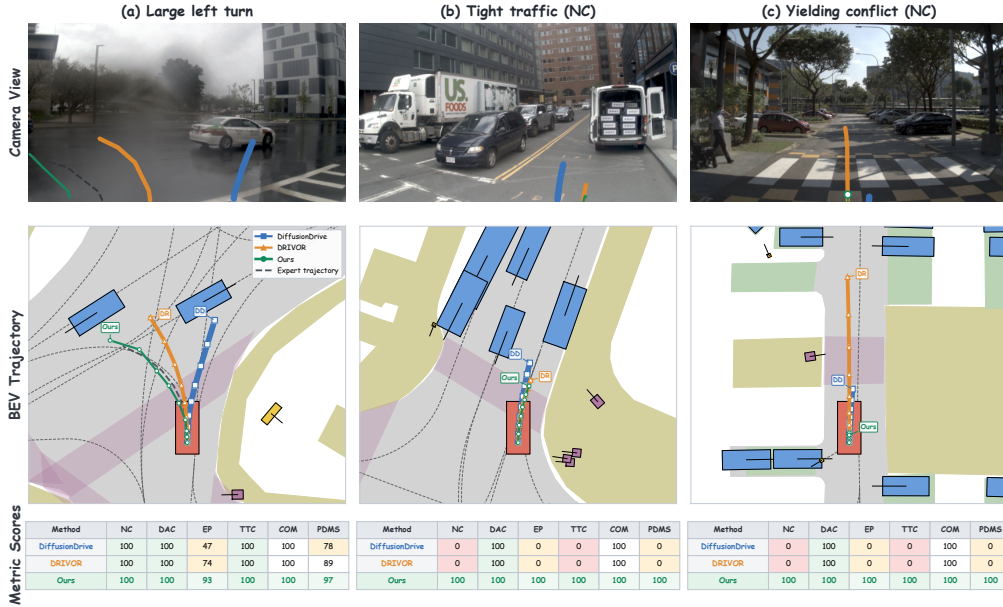


Figure 4: **Qualitative comparison of trajectory selection.** Camera views (top), BEV trajectories (middle), and per-scene metric scores (bottom) are shown for (a) a large left turn, (b) tight traffic, and (c) a yielding conflict. The trajectories produced by DiffusionDrive, DrivoR, and DA-WAM are shown in blue, orange, and green, respectively, with the expert trajectory shown as a dashed line. In (a), DA-WAM more closely follows the expert trajectory and achieves the highest EP and PDMS scores. In (b) and (c), DA-WAM avoids the conflicts that lead to NC and TTC failures for both baselines.

Candidate-Count Ablation. Table 5 evaluates sensitivity to the number of trajectory candidates while keeping the remaining configuration fixed. PDMS improves consistently up to 32 candidates and remains close with 64 candidates, supporting the use of 32 candidates in the final configuration.

Table 5: **Candidate-count ablation.** Influence of the number of candidate trajectories on the NAVSIM-v1 navtest split, with all other settings fixed. PDMS is scaled by 100, and higher is better.

Candidates	1	8	16	32	64
PDMS	87.11	90.76	91.89	93.68	93.68

4.4 QUALITATIVE ANALYSIS

Fig. 4 compares DA-WAM with DiffusionDrive and DrivoR across three representative driving scenarios. In the large-left-turn scenario, all methods remain collision-free, whereas DA-WAM more closely follows the expert trajectory and achieves substantially higher EP and PDMS scores. In the tight-traffic and yielding-conflict scenarios, both baselines incur NC and TTC failures, while DA-WAM selects safer trajectories that avoid the conflicting agents and attain full scores across all reported metrics. These examples demonstrate that the proposed scorer maintains progress when executing an unconstrained turn while prioritizing safety in the presence of imminent traffic conflicts.

5 CONCLUSION

World models are most valuable for autonomous driving when their predictions directly inform trajectory selection. DA-WAM closes the gap between prediction and planning by learning future representations together with the driving task, predicting a distinct future for each candidate trajectory, and evaluating every candidate against its corresponding outcome. Expert-matched supervision keeps future learning consistent with the observed data, while safety-critical hard negatives improve discrimination near planning boundaries. Experiments on NAVSIM-v1 and NAVSIM-v2 achieve competitive results, and ablation studies provide evidence for the contribution of the main design choices. Overall, our findings suggest that predicting a plausible future alone may not be sufficient for effective planning. The prediction should also correspond to the action being evaluated and contribute to the final decision. By strengthening this connection, DA-WAM offers a practical approach to making future modeling more relevant to trajectory planning.

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A NOTATION

Table 6 summarizes the main symbols used in the formulation of DA-WAM.

Table 6: Main notation used in DA-WAM.

Symbol	Meaning
$X_t, X_{t+\Delta}$	Current and observed future visual inputs.
$\mathcal{T} = \{\tau_i\}_{i=1}^N$	Set of N candidate ego trajectories.
$\tau^{\text{exp}}, \tau_j^-, \tau^*$	Expert, hard-negative, and finally selected trajectories.
$E_\theta, E_{\bar{\theta}}$	Online encoder and EMA target encoder.
$Z_t, Z_{t+\Delta}$	Current scene latent and observed future latent target.
$a_i = E_\tau(\tau_i)$	Action representation of candidate τ_i .
$\hat{Z}_i = P_\phi(Z_t, a_i)$	Future latent predicted for candidate τ_i .
S_ψ	Shared future-latent-conditioned trajectory scorer.
i^{exp}	Index of the candidate matched to the expert trajectory.
$\mathbf{q}_i, \hat{\mathbf{q}}_i$	Target and predicted planning-factor vectors.
$s_i, \hat{s}_i$	Target and predicted overall trajectory utilities.
M, D	Number of latent tokens and token dimension.
μ	Momentum coefficient for the EMA target encoder.
$\mathcal{K}$	Set of supervised planning factors.
$\mathcal{L}_{\text{pred}}, \mathcal{L}_{\text{factor}}$	Future-prediction and planning-factor losses.
$\mathcal{L}_{\text{score}}, \mathcal{L}_{\text{rank}}$	Utility-regression and pairwise-ranking losses.
λ	Weights used to combine the training losses.